

TSETSERLEG, Mongolia

WMO: 442820

Lat: 47.4717N

Lon: 101.4628E

Elev: 1691

StdP: 82.59

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.0	-25.5	-33.8	0.2	-25.5	-31.9	0.2	-23.3	10.2	-8.3	9.3	-10.0	0.8	320	0.654

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	27.1	13.7	25.1	13.2	23.5	12.6	16.1	23.0	15.0	21.8	14.1	20.7	3.1	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.8	12.1	18.2	12.6	11.2	17.4	11.5	10.4	16.8	51.4	23.1	48.2	21.8	45.4	20.8	21.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
9.5	8.2	7.0	DB	-29.7	30.4	3.2	1.9	-32.0	31.8	-33.9	32.9	-35.7	34.0	-38.1	35.4	(4)
			WB	-29.9	18.1	3.2	1.7	-32.2	19.3	-34.0	20.3	-35.8	21.2	-38.2	22.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.7	-15.0	-11.7	-4.8	3.5	9.2	14.4	16.6	14.8	9.4	1.7	-6.6	-12.4
	DBStd	12.03	5.63	5.93	6.28	5.16	4.84	3.56	3.00	3.05	4.34	4.96	6.06	5.27
	HDD10.0	3637	775	608	459	201	74	7	1	3	60	257	498	693
	HDD18.3	6113	1033	842	717	446	285	123	71	115	267	515	748	952
	CDD10.0	592	0	0	0	5	49	139	203	152	43	1	0	0
	CDD18.3	28	0	0	0	0	1	6	16	6	0	0	0	0
	CDH23.3	407	0	0	0	0	24	108	181	84	11	0	0	0
	CDH26.7	77	0	0	0	0	2	20	41	14	1	0	0	0
Wind	WSAvg	2.5	2.0	2.2	2.8	3.3	3.1	2.4	2.0	2.1	2.4	2.4	2.5	2.2
Precipitation	PrecAvg	329	2	3	7	16	33	64	83	75	24	14	6	3
	PrecMax	480	11	14	16	40	80	161	153	189	66	50	14	23
	PrecMin	155	0	0	0	0	2	10	15	2	0	0	0	0
	PrecStd	70	2	3	4	8	19	30	36	40	17	11	4	4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.7	4.6	13.4	20.4	26.2	29.2	30.7	28.6	24.8	17.5	10.4	2.1
		MCWB	-3.1	-1.5	3.6	7.8	10.6	13.8	15.8	14.3	11.6	6.3	2.5	-2.8
	2%	DB	-1.5	2.0	10.0	17.2	23.1	26.2	27.4	25.6	21.9	14.7	7.1	-0.7
		MCWB	-5.4	-3.2	1.8	6.0	10.0	12.6	14.8	13.6	10.4	5.2	0.7	-4.7
	5%	DB	-4.2	-0.4	7.4	14.9	20.8	24.1	25.1	23.5	19.7	12.3	4.5	-2.7
		MCWB	-7.5	-4.7	0.2	5.1	8.9	12.1	14.3	13.1	9.3	4.1	-1.0	-6.2
	10%	DB	-6.8	-2.7	4.8	12.6	18.6	22.3	23.3	21.6	17.5	9.9	2.0	-4.9
		MCWB	-9.4	-6.3	-1.3	3.8	7.9	11.7	13.8	12.6	8.4	2.8	-2.6	-7.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.9	-1.4	4.1	8.4	13.2	15.8	18.5	17.2	13.1	7.2	3.1	-2.6
		MCDB	1.3	4.2	12.8	18.5	22.5	24.3	25.6	23.4	20.0	15.6	9.4	1.7
	2%	WB	-5.4	-3.1	1.9	6.6	10.9	14.4	16.6	15.5	11.6	5.8	0.8	-4.7
		MCDB	-1.7	1.7	9.3	15.9	20.8	22.2	23.5	21.2	18.9	13.6	6.8	-0.9
	5%	WB	-7.4	-4.7	0.3	5.4	9.6	13.5	15.6	14.4	10.4	4.4	-0.8	-6.2
		MCDB	-4.3	-0.5	6.8	14.1	19.5	20.8	22.6	20.3	17.7	11.5	4.1	-2.9
	10%	WB	-9.4	-6.3	-1.1	4.1	8.3	12.6	14.6	13.5	9.1	3.1	-2.4	-7.9
		MCDB	-6.8	-2.7	4.6	12.3	17.5	20.0	21.4	19.6	16.3	9.7	1.8	-5.1
Mean Daily Temperature Range	5% DB	MDBR	10.1	11.5	11.7	11.4	11.9	11.6	10.7	11.0	12.0	11.3	10.3	9.6
		MCDBR	12.7	13.9	14.1	14.7	15.6	15.2	14.3	14.5	15.2	14.7	12.9	11.6
		MCWBR	9.8	10.2	8.7	7.9	7.6	6.4	5.6	6.0	7.1	8.1	8.4	8.8
	5% WB	MCDBR	12.5	13.8	13.6	13.9	14.6	12.4	11.8	11.5	13.2	14.1	12.2	11.5
		MCWBR	9.7	10.3	8.6	7.8	7.5	6.0	5.6	5.9	6.7	8.2	8.2	8.8
Clear-Sky Solar Irradiance	taub	0.227	0.252	0.297	0.336	0.349	0.357	0.349	0.328	0.296	0.273	0.248	0.230	
	taud	2.398	2.357	2.297	2.270	2.283	2.364	2.434	2.461	2.500	2.482	2.428	2.380	
	Ebn at Noon	892	931	926	916	912	905	908	916	922	894	852	839	
	Edh at Noon	65	86	108	124	127	118	109	101	88	76	63	59	
All-Sky Solar Radiation	RadAvg	1.93	2.99	4.47	5.68	6.33	6.23	5.91	5.34	4.60	3.16	2.01	1.55	
	RadStd	0.07	0.10	0.15	0.25	0.42	0.35	0.36	0.27	0.22	0.16	0.11	0.05	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (6 neighbors)	N/A	N/A	-2.59	N/A	-0.78	-1.00	N/A	N/A	N/A	N/A

Nomenclature: See separate page